

Homework 9: Model selection

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Problem 22: d-separation

Part i: All variables d-separated from A given {C,E}

No nodes are d-separated from A given $\{C, E\}$, brief explanations are provided below:

B is not because $A \rightarrow B$ is an open chain.

C does not apply because it is given.

D is not because $A \rightarrow B \rightarrow D$ is an open chain.

F is not because $A \rightarrow B \rightarrow F$ is an open chain.

G has two paths that come from A : $A \rightarrow C \rightarrow G$, is closed because C is observed, but $A \rightarrow B \rightarrow D \rightarrow G$ is an open chain, so G is also not d-separated.

E does not apply because it is given.

Part ii: True or false independence statements

Part i:

C is independent of D : False.

There are two paths going from C to D . $C \rightarrow G \leftarrow D$ where G is a collider is closed because G is not given. $C \leftarrow A \rightarrow B \rightarrow D$. Here there is an open fork $C \leftarrow A \rightarrow B$ and an open chain $A \rightarrow B \rightarrow D$, so information flows and C and D are not independent.

Part ii:

B is conditionally independent of C given $\{F, G\}$: False.

There are three paths going from B to C : $B \leftarrow A \rightarrow C$, $B \rightarrow D \rightarrow G \leftarrow C$ and $B \rightarrow F \rightarrow E \leftarrow D \rightarrow G \leftarrow C$. $B \leftarrow A \rightarrow C$ is an open fork, so information flows. This is sufficient to state they are not conditionally independent. $B \rightarrow D \rightarrow G \leftarrow C$ would also be open because $B \rightarrow D \rightarrow G$ is a chain and $D \rightarrow G \leftarrow C$ is open because G is a collider (normally closed) but because it is given it opens. $B \rightarrow F \rightarrow E \leftarrow D \rightarrow G \leftarrow C$ would be closed because both $B \rightarrow F \rightarrow E$ is a chain but F is given, so it is closed, and also $F \rightarrow E \leftarrow D$ with E as a collider, which is not given, so is closed.

Part iii:

G is conditionally independent of E given $\{A, D\}$: True.

There would be four paths that connect G and E , all 4 are blocked because firstly $C \leftarrow A \rightarrow B$ is a fork with A given, closing the two paths passing through here. Secondly $G \leftarrow D \rightarrow E$ is a fork and $G \leftarrow D \leftarrow B$ where D is given, so they close and no path can reach E , making G and E independent given $\{A, D\}$.

Part iv:

C is conditionally independent of F given $\{G, E\}$: False.

$C \rightarrow G \leftarrow D$ and $D \rightarrow E \leftarrow F$ are the only two v structures in the graph with G and E as colliders. By observing G and E these colliders are open and the entire Bayesian network is open, so C and F are not conditionally independent. There is also another, “easier” path, $C \leftarrow A \rightarrow B \rightarrow F$ which is open because fork $C \leftarrow A \rightarrow B$ is open and chain $A \rightarrow B \rightarrow F$ is also open.

Part v:

C is conditionally independent of E given the Markov blanket of C : True.

The Markov blanket of a node in a Bayesian network consists of three types of neighbours: parents, children, and parents of its children. So for C , $MB(C)$: $\{A, G, D\}$ because A is its parent, G is its children, and D is the parent of its children, G . Observing A blocks the fork $C \leftarrow A \rightarrow B$, blocking any path through A . Observing G opens the collider G in $C \rightarrow G \leftarrow D$, however, observing D , blocks both the chain $G \leftarrow D \leftarrow B$ and the fork $G \leftarrow D \rightarrow E$. Since all paths between C and E pass through either A or D , they are all blocked, meaning the two nodes are conditionally independent.

Problem 23: Structure learning in R

```
# loading the data
simulated_DAG <- read.csv("simulated_DAG.csv")
```

Next, the different claims of problem 22 part i will be evaluated. Since C and E are given, they are not evaluated.

```
# A is conditionally independent of B (d-separated) given {C,E}
A_CE_results <- lm(A ~ C+E, simulated_DAG)
B_CE_results <- lm(B ~ C+E, simulated_DAG)
A_B_CE_results <- cor.test(A_CE_results$residuals,B_CE_results$residuals)
print(A_B_CE_results)
```

Pearson's product-moment correlation

```
data: A_CE_results$residuals and B_CE_results$residuals
t = 15.607, df = 998, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.3916990 0.4914352
sample estimates:
      cor
0.4429364
```

```
# A is conditionally independent of D (d-separated) given {C,E}
D_CE_results <- lm(D ~ C+E, simulated_DAG)
A_D_CE_results <- cor.test(A_CE_results$residuals,D_CE_results$residuals)
print(A_D_CE_results)
```

Pearson's product-moment correlation

```
data: A_CE_results$residuals and D_CE_results$residuals
t = 5.3793, df = 998, p-value = 9.317e-08
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1069823 0.2274878
sample estimates:
```

```
cor
0.1678621
```

```
# A is conditionally independent of F (d-separated) given {C,E}
F_CE_results <- lm(F ~ C+E, simulated_DAG)
A_F_CE_results <- cor.test(A_CE_results$residuals,F_CE_results$residuals)
print(A_F_CE_results)
```

Pearson's product-moment correlation

```
data: A_CE_results$residuals and F_CE_results$residuals
t = 8.9332, df = 998, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.2137177 0.3285564
sample estimates:
      cor
0.2721056
```

```
# A is conditionally independent of G (d-separated) given {C,E}
G_CE_results <- lm(G ~ C+E, simulated_DAG)
A_G_CE_results <- cor.test(A_CE_results$residuals,G_CE_results$residuals)
print(A_G_CE_results)
```

Pearson's product-moment correlation

```
data: A_CE_results$residuals and G_CE_results$residuals
t = 2.9198, df = 998, p-value = 0.003581
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.03021167 0.15315168
sample estimates:
      cor
0.09203239
```

All p-values for the 4 tested variables: B , D , F and G are lower than 0.05, the null hypothesis of no correlation, or of conditional independence, or of d-separation can be rejected. Hence, as suggested from the graph structure in problem 22, there are no d-separated variables with A given $\{C, E\}$.

Next, the different claims of problem 22 part ii will be evaluated.

```
# C independent of D
C_D_results <- cor.test(simulated_DAG$C,simulated_DAG$D)
print(C_D_results)
```

Pearson's product-moment correlation

```
data: simulated_DAG$C and simulated_DAG$D
t = 6.7497, df = 998, p-value = 2.508e-11
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1488781 0.2674713
sample estimates:
      cor
0.2089428
```

A p-value in the order of 10^{-11} indicates that the null hypothesis of no correlation can be rejected, and therefore, C is not marginally independent of D as previously suggested based on the graph structure in problem 22.

```
# B is conditionally independent of C given {F,G}
B_FG_results <- lm(B ~ F+G, simulated_DAG)
C_FG_results <- lm(C ~ F+G, simulated_DAG)
B_C_FG_results <- cor.test(B_FG_results$residuals,C_FG_results$residuals)
print(B_C_FG_results)
```

Pearson's product-moment correlation

```
data: B_FG_results$residuals and C_FG_results$residuals
t = 0.062881, df = 998, p-value = 0.9499
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
-0.06001002 0.06397565
sample estimates:
      cor
0.001990465
```

A p-value of 0.95 indicates that there is no correlation, and therefore, B is conditionally independent of C given $\{F, G\}$. This is surprising, based on the graph structure, in problem 22 it was proposed that given the evidence, B and C were not conditionally independent. An explanation for this is that the paths effect cancels out, and when viewing the data the nodes appear to be uncorrelated. There are two active paths between B and C given $\{F, G\}$: $B \leftarrow A \rightarrow C$ and $B \rightarrow D \rightarrow G \leftarrow C$. It could be that the correlation for one path is positive, and for the other negative, cancelling each other out. Infact, while d-separation guarantees independence, d-connection (what the graph would suggest) only implies potential dependence.

```
# G is conditionally independent of E given {A,D}
G_AD_results <- lm(G ~ A+D, simulated_DAG)
E_AD_results <- lm(E ~ A+D, simulated_DAG)
GE_A_D_results <- cor.test(G_AD_results$residuals,E_AD_results$residuals)
print(GE_A_D_results)
```

Pearson's product-moment correlation

```
data:  G_AD_results$residuals and E_AD_results$residuals
t = -0.13867, df = 998, p-value = 0.8897
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.06636460  0.05761919
sample estimates:
      cor
-0.004389576
```

A p-value of 0.89 indicates that there was no significant correlation between the residuals, and therefore, G is conditionally independent of E given $\{A, D\}$ as previously suggested based on the graph structure in problem 22.

```
# C is conditionally independent of F given {G,E}
C_GE_results <- lm(C ~ G+E, simulated_DAG)
F_GE_results <- lm(F ~ G+E, simulated_DAG)
C_F_GE_results <- cor.test(C_GE_results$residuals,F_GE_results$residuals)
print(C_F_GE_results)
```

Pearson's product-moment correlation

```

data: C_GE_results$residuals and F_GE_results$residuals
t = 5.9685, df = 998, p-value = 3.323e-09
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1250920 0.2448209
sample estimates:
      cor
0.1856454

```

A p-value in the order of 10^{-9} indicates that the null hypothesis of no correlation can be rejected, and therefore, C is not conditionally independent of F given $\{G, E\}$ as previously suggested based on the graph structure in problem 22.

```

# C is conditionally independent of E given {A,G,D} (the Markov blanket of C)
C_AGD_results <- lm(C~A+G+D, simulated_DAG)
E_AGD_results <- lm(E~A+G+D, simulated_DAG)
C_E_AGD_results <- cor.test(C_AGD_results$residuals,E_AGD_results$residuals)
print(C_E_AGD_results)

```

Pearson's product-moment correlation

```

data: C_AGD_results$residuals and E_AGD_results$residuals
t = 0.49763, df = 998, p-value = 0.6189
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.04628794 0.07766758
sample estimates:
      cor
0.01575034

```

A p-value of 0.62 indicates that there was no significant correlation, and therefore, C is conditionally independent of E given $\{A, G, D\}$ as previously suggested based on the graph structure in problem 22.

Part ii: PC algorithm

```

suppressPackageStartupMessages(library(pcalg))

```

Warning: package 'pcalg' was built under R version 4.5.3

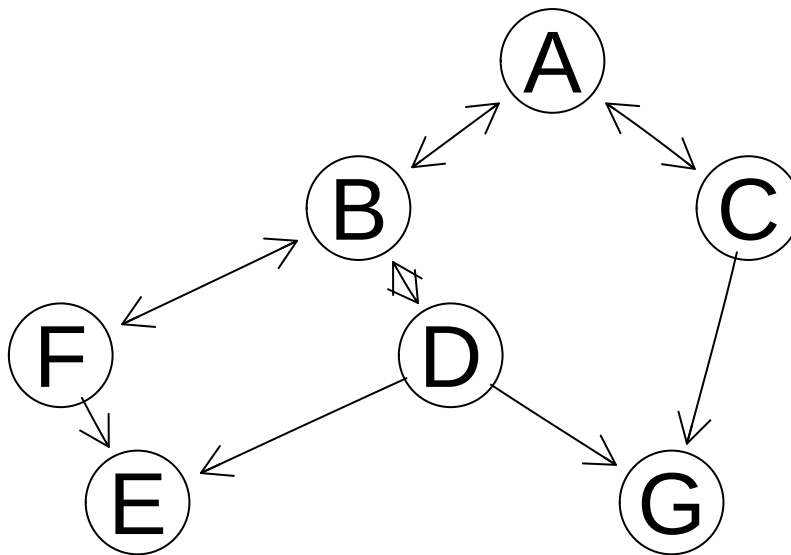
```
cor_mat <- cor(simulated_DAG)
n <- nrow(simulated_DAG)
suffStat <- list(C = cor_mat, n = n)

pc.fit <- pc(suffStat = suffStat, alpha = 0.05, indepTest = gaussCItest, labels = colnames(suffStat))

plot(pc.fit, main = "Estimated CPDAG")
```

Loading required namespace: Rgraphviz

Estimated CPDAG



Part iii: MCMC with BGe Score

```
suppressPackageStartupMessages(library(BiDAG))
```

Warning: package 'BiDAG' was built under R version 4.5.3

```
Score <- scoreparameters("bge",simulated_DAG)
maxBM <- learnBN(Score, algorithm = "orderIter")
print(maxBM$CPDAG)
```

```
7 x 7 sparse Matrix of class "dgCMatrix"
  A B C D E F G
A . 1 1 . . . .
B 1 . . 1 . 1 .
C 1 . . . . . 1
D . 1 . . 1 . 1
E . . . . . . .
F . 1 . . 1 . .
G . . . . . . .
```

Or better plotting the graph with edges:

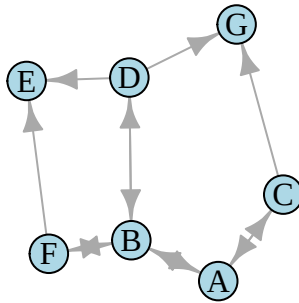
```
suppressPackageStartupMessages(library(igraph))
```

Warning: package 'igraph' was built under R version 4.5.3

```
cpdag_graph <- graph_from_adjacency_matrix(maxBM$CPDAG)

# 2. Plot the graph!
plot(cpdag_graph,
     main = "Completed partially DAG (CPDAG) from orderIter MCMC",
     vertex.color = "lightblue",
     vertex.size = 30,
     vertex.label.color = "black",
     edge.arrow.size = 0.8)
```

Completed partially DAG (CPDAG) from orderlter MCMC



Part iv: Comparison of predicted DAG versus PC algorithm and MCMC with BGe score estimates

Both the PC algorithm and the MCMC with BGe score estimates suggest identical graphs to that of problem 22!